

Juneha Hwang

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Education

Korea Advanced Institute of Science and Technology (KAIST)

August 2023-

Bachelor of Science in Computer Science (GPA: 4.01 / 4.30)

Daejeon, South Korea

- **Relevant Coursework:** Linear Algebra for Data Science (Python), Programming Principles (Rust), Data Structures (Java), Introduction to Algorithms, Machine Learning, Systems Programming (C), Programming Languages (Scala)

Experience

Linq Alpha

Jan 2025–Present

Data Engineer

Seoul, South Korea

- Architected a FastAPI-based API layer and asynchronous processing backend (Celery/Redis) to preprocess and deliver large public/licensed datasets with observability for long-running jobs.
- Managed and optimized large-scale financial datasets using PostgreSQL and Alembic, ensuring data integrity and performance.
- Engineered an entity resolution pipeline that maps ambiguous company names (e.g. "MS", "Apple") to canonical stock identifiers using a multi-stage system (cache → trigram search on DB → vector DB search → Decision-making using LLM).
- Designed and deployed a semantic document chunking system that transforms unstructured financial documents (earnings calls, broker research, internal investment notes, etc.) into structured queryable metadata, processing variable-format PDFs through a multi-batch LLM pipeline with multi-layer JSON parsing to handle non-deterministic LLM model outputs.

FBA Quant (Quantitative Finance Research Group)

Feb 2024–Present

Quant Developer Track

Seoul, South Korea

- Developed expertise in asset pricing, portfolio management, and foundational financial engineering.
- Engineered an algorithmic trading system with the Interactive Brokers API, utilizing parallel processing and comprehensive backtesting in Python.

Projects

FABRIC Interpreter & Type Checker (KAIST CS320 Project #3) | *Scala*

- Implemented a Scala interpreter + static type checker for FABRIC, a functional language with polymorphic ADTs and pattern matching.
- Added a desugaring pass from concrete syntax to a simpler AST to reduce interpreter complexity.

Facial Emotion Recognition Program | *Keras, Tensorflow, Scikit-Learn*

- Designed and implemented a Convolutional Neural Network (CNN) using Keras to classify facial emotions into seven distinct categories. Improved accuracy from 60% to 70% with hyperparameter-tuning.
- Developed a real-time application in Python integrating a live camera feed to identify and categorize facial emotions, leveraging machine learning models for enhanced human interaction.

Certificates and Honors

Certificates

- Machine Learning with Python: A Practical Introduction, Deep Learning with Tensorflow, Deep Learning with Tensorflow (*IBM*)
- Using Python for Research (*HarvardX*)

Honors

- Samsung Collegiate Programming Cup (SCPC) 2024, 2025 Finalist
- Union of Clubs for Programming Contests (UCPC) 2024, 2025 Finalist
- Fall 2023 Dean's List (KAIST)

Technical Skills

Programming Languages: Python, C / C++, Rust, Scala, JavaScript, Typescript, Ruby, SQL, HTML/CSS

Tech Stack: FastAPI, PostgreSQL, Celery, Redis, React, Scikit-Learn, Numpy, Pandas, Git

Languages: Korean (native), English (native), Japanese (conversational)